

Robert G. Gambee

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AI Research Engineer dedicated to improving the world for future generations

Core Competencies

- Proven ability to rapidly master new technologies and adapt in response to evolving needs
- Expertise at driving complex projects from inception to production
- Talent for balancing big-picture strategic thinking and detail-oriented execution
- Experience designing systems to be reliable, scalable and maintainable
- Passion for leveraging technical skills to create meaningful positive impact

Software Skills

Proficient

- Python
- JavaScript, TypeScript, React
- SQL, PostgreSQL, BigQuery
- Git, Jira
- Claude Code, Cursor

Experienced

- PyTorch, scikit-learn
- Docker
- GitHub Actions
- Langfuse, OpenTelemetry
- Bash

Familiar

- Google Cloud Platform
- Amazon Web Services
- Next.js
- Go
- Rust

Professional Experience

FutureSearch: Startup using AI for research and forecasting 2025 to Present
AI Research Engineer

Responsibilities

- Evaluate and enhance capabilities of AI research agents
- Build robust pipelines for running thousands of AI agents concurrently
- Work throughout the stack to rapidly iterate on product ideas
- Log data and build dashboards to monitor performance, stability and cost
- Keep up to date with latest developments in AI capabilities and tools
- Communicate with users to get their feedback and help them get the most out of our product

Technical Projects

- Tech Lead for forecasting research 2026 to Present
 - Led three-person effort to improve the accuracy of our state-of-the-art forecaster
 - Implemented system of thousands of self-reinforcing forecasts that continually improves over time
 - Built upon three-tier sharing system to protect user privacy
- Tech Lead for tools to let AI research agents search the internet and read documents 2025 to Present
 - Developed and evaluated tools to answer queries about web pages
 - Extensively analyzed agent transcripts to identify failure modes
 - Eliminated leading failure mode by switching to superior page retrieval service
 - Optimized page reading flow to reduce maximum latency by factor of 5
- Open-source library for interfacing with our services 2026 to Present
 - Designed API, SDK and MCP server to be easy to use yet powerful
 - Managed process for publishing releases
- AI validator for uncertain claims 2025
 - Developed workflow to search the internet for relevant information
 - Created judge to weigh evidence and estimate the probability that the claim is true
 - Wrote and maintained a suite of challenging evaluations
 - Benchmarked custom validator against other approaches across many models

MIT FutureTech – AI Risk Initiative

2026 to Present

Research Engineer (Part-Time Contractor)

Technical Projects

- Python library for building document processing pipelines 2026 to Present
 - Created suite of reusable components to accelerate building of future pipelines
 - Wrote extensive unit tests, automated checks and AI guidelines to promote correctness and consistency
- Automated pipeline for collecting and classifying AI risks 2026 to Present
 - Designed and built pipeline to process over 30,000 documents far faster than previous manual one
 - Compiled human-labeled validation datasets in collaboration with other researchers
 - Crafted LLM prompts to screen documents, balancing false positives against false negatives
 - Engineered retry and resume features to make pipeline robust to errors
 - Optimized pipeline to balance accuracy, cost and speed
- Interactive explorer for understanding the initiative's many pipelines 2026 to Present
 - Created graph view to visualize connections between dozens of pipeline stages
 - Included details about each pipeline stage to provide user with in-depth understanding

Formlabs: Industry leader in professional 3D printing

2015 to 2025

Systems Integration Engineer III

Responsibilities

- Architected software which was maintainable, scalable and testable
- Owned key printer systems for the entire product cycle, driving them from inception to public release
- Understood complex interactions between printer systems, keeping both details and big picture in mind
- Optimized for printer reliability through robust design and failure mode prediction
- Analyzed and visualized printer data to answer pressing questions and inform business decisions
- Mentored junior team members to foster their technical abilities

Technical Projects

- Senior embedded developer for Form 4 and Form 4L 2021 to 2025
 - Thoughtfully crafted powerful yet understandable API to control all aspects of product functionality
 - Comprehensively audited API for security vulnerabilities and coordinated plan to address them
 - Advised architectural decisions for embedded and desktop software
- Data management on user-replaceable components for Form 4 and Form 4L 2023 to 2024
 - Designed a unified architecture for all components, agnostic to interface and data format
 - Wrote extensive validation checks to be robust to failures when reading or writing
 - Secured system against counterfeiting to protect company's primary revenue stream
 - Thoroughly tested all code with automated checks
- Print preparation routine for Form 4 and Form 4L 2021 to 2024
 - Sped up routine by a factor of 5 to 10 compared to previous product, vastly improving user experience
 - Wrote predictive checks to give user advance warning of issues and avoid interrupting prints
 - Implemented specification for how to handle over 50 possible errors
- Dashboard for plotting live sensor data 2021
 - Independently developed over four days during company hackathon
 - Wrote backend in Go, wrote frontend in JavaScript, streamed data via WebSockets
 - Recognized by the CEO in a company-wide email as one of the most impressive projects that year

Achievements

- Recipient of Formlabs' Perform Award, which recognizes top 10% of employees 2020 and 2023

Volunteer Experience

AI Governance and Safety Canada

2024

- Designed and implemented flexible system for scraping information relevant to the AI safety community
- Leveraged LLM to robustly scrape many sites with zero site-specific tuning required
- Initially developed to find upcoming events, but can be extended to other types of content, e.g. publications
- Created automated workflow using GitHub Actions to run scraper and publish output to database

Independent AI Research

Do LLMs have stable preferences?

2025 to 2026

- Built upon findings in Claude 4 System card to understand how preferences vary under different conditions
- Used pairwise comparisons between tasks to determine the LLM's implicit rating for each task
- Compared multiple models, response formats, orders and numbers of tasks
- Found that ratings for individual tasks combine coherently when considering sequences of tasks
- Uncovered strong dependency on response format: free-form vs. structured output
- Results have implications for AI welfare, highlighting challenges with satisfying LLM preferences

Reproduction of “The Capacity for Moral Self-Correction in Large Language Models”

2023

- Loaded and processed tens of thousands of samples from three different datasets
- Submitted API requests asynchronously, with automatic retries and rate limiting
- Analyzed bias in model responses according to three different metrics
- Compared and contrasted results to demonstrate influence of RLHF training vs. prompt engineering

Personal Projects

Knight Grid

2026

Visualizer for complex, emergent patterns when placing chess pieces according to simple rules

- Utilized Three.js and custom shaders to efficiently display hundreds of millions of points
- Optimized algorithm to place 1 million pieces per second by making it $O(N)$ instead of naive $O(N^2)$
- Encoded configuration in URL for easy sharing

Chronicle

2023

Web app to keep track of how one spends one's time

- Used Django framework to manage HTTP requests and access SQLite database
- Presented data as a table for sorting and filtering, as well multiple charts for visualization
- Set up automated test and deployment workflows using GitHub Actions

SCAFFOLD

2023

Completed as part of AI Safety Camp (3 person team plus advisor)

- Built React web app to generate feedback on one's research ideas using GPT
- Fine tuned model to make its responses more relevant to AI safety research

Publications

Deep Research Bench: Evaluating AI Web Research Agents (2025). FutureSearch: Nikos I. Bosse, Jon Evans, **Robert G. Gambee**, Daniel Hnyk, Peter Mühlbacher, Lawrence Phillips, Dan Schwarz, Jack Wildman. <https://doi.org/10.48550/arXiv.2506.06287>

Barium titanate nanoparticles: Short-range lattice distortions with long-range cubic order (2018). Chenyang Shi, Simon J. L. Billinge, Eric Puma, Sun Hwi Bang, Nathaniel J. H. Bean, Jean-Claude de Sugny, **Robert G. Gambee**, Richard C. Haskell, Adrian Hightower, Todd C. Monson. <https://doi.org/10.1103/PhysRevB.98.085421>

Permittivity effects of particle agglomeration in ferroelectric ceramic-epoxy composites using finite element modeling (2018). Jonas L. Kaufman, Scott H. Tan, Kirkclann Lau, Ashka Shah, **Robert G. Gambee**, Chris Gage, Lupe MacIntosh, Albert Dato, Peter N. Saeta, Richard C. Haskell, Todd C. Monson. <https://doi.org/10.1063/1.5053442>

Education

Harvey Mudd College, Claremont, CA

Bachelor of Science in Engineering with High Distinction

2011 to 2015

- GPA: 3.8
- Inducted into Tau Beta Pi, national engineering honor society
- Recognized on Dean's List of top performing students

2014

2012 to 2015

Undergraduate Projects

SpaceX, Hawthorne, CA & Harvey Mudd College

2014 to 2015

Recoverable Flight Data Recorder (5 person team)

- Designed housing and selected materials to protect electronics from rocket explosion
- Built and tested prototypes according to SMC-S-016 and other specifications
- Contributed to software for receiving flight data over UDP and saving to SD card

Academic Research, Harvey Mudd College

2014 to 2015

Gas Permeation Across Nanocomposite Polymer Membranes (5 to 8 person team)

- Performed gas permeation experiments on synthesized membranes
- Ran and analyzed molecular simulations containing over ten thousand atoms each
- Wrote grant proposal for Amazon EC2 resources that decreased runtime by an order of magnitude

Sandia National Laboratories, Albuquerque, NM & Harvey Mudd College

2013 to 2014

Measurement of Barium Titanate Nanoparticle Permittivity (5 person team)

- Developed analytical and numerical models for interpretation of experimental data
- Presented work at Materials Research Society conference as an invited speaker
- Project findings were later published in American Institute of Physics Advances